

Implementing Deep Learning in Estimation of Heterogeneous Taste Parameters in Hierarchical Structural Models

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Abstract

This paper introduces NN+S, a novel semi-parametric framework that integrates neural networks with structural models to estimate heterogeneous taste parameters with enhanced accuracy and interpretability, offering significant applications in marketing. By designing a neural network architecture that models both observed and unobserved heterogeneity, as well as intra-individual random taste shocks, NN+S captures complex preference structures while retaining structural interpretability. The NN+S model enables companies to predict consumer responses to counterfactual scenarios, such as new product introductions, personalized pricing, or changes in product attributes. Monte Carlo simulations assess NN+S’s ability to recover the true taste parameters. Empirical results show that NN+S outperforms linear hierarchical Bayesian models (LHBM) and the model by [Farrell et al. \(2020\)](#) (FLM) on a classical margarine purchase dataset for the same individuals, but new purchases (in-sample predictions), leveraging purchase histories for precise targeting and market segmentation. For new customers (out-of-sample predictions), NN+S achieves a slightly higher hit rate than alternative approaches. Scalable via standard neural network tools, NN+S empowers marketers to optimize strategies such as dynamic pricing and product recommendations, providing a flexible and interpretable tool for demand estimation.

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1 Introduction

Understanding heterogeneous consumer preferences is fundamental to economic modeling, marketing strategy, and behavioral science. Structural models offer interpretable parameters that capture observed heterogeneity (e.g., demographic effects) and unobserved heterogeneity (e.g., latent traits), as well as taste shocks; yet, they often struggle with complex, non-linear relationships due to the rigidity of their functional forms. Neural networks (NNs), while effective at learning intricate relationships, lack the structural interpretability required for causal inference, such as counterfactual analysis in marketing applications. This paper introduces NN+S, a novel semi-parametric framework that combines the flexibility of neural networks with the interpretability of structural models to estimate individual taste parameters with enhanced accuracy, providing significant advantages for marketing practitioners.

NN+S innovates by integrating key interpretable structural components into the design of neural networks. The components (or parts) of NN+S account for: observed heterogeneity through linear (Δz_i) and non-linear ($G(z_i)$) demographic effects, unobserved heterogeneity via individual uncorrelated variation (u_i) and correlated variation through segment weights (Av_i), and intra-individual random taste shocks (μ^m). This design allows NN+S to capture complex preference structures while preserving the ability to interpret parameters for actionable insights. For instance, firms can leverage NN+S to estimate individual price sensitivities for personalized pricing, predict product choices for targeted recommendations, or segment markets based on inferred preference groups, utilizing both observed demographics and unobserved traits derived from purchase histories.

Our approach bridges a critical gap in the literature. While Farrell et al. (2020) explored combining neural networks with structural models for observed heterogeneity, they offered limited practical guidance for addressing unobserved heterogeneity or non-IIA behavior. NN+S provides detailed implementation strategies, leveraging standard NN tools (e.g., Keras, TensorFlow) for scalability and GPU acceleration. Monte Carlo simulations assess NN+S’s ability to effectively estimate taste parameters, particularly in settings with

correlated unobserved heterogeneity.

Empirical analysis on a margarine purchase dataset (Allenby and Rossi, 1991) demonstrates that NN+S outperforms linear hierarchical Bayesian models (LHBM) and the model by Farrell et al. (2020) (FLM) for in-sample predictions (63% vs. 57% and 47% hit rates, respectively), capitalizing on purchase histories to enhance individual-specific effects. For out-of-sample predictions, NN+S (45.1%) slightly outperforms LHBM (44.4%), tightly followed by FLM (44.3%).

We believe that NN+S is particularly valuable for marketing applications where firms have access to detailed consumer data. For example, a retailer can use NN+S to predict demand for a new product by modeling how unobserved heterogeneity (e.g., latent health consciousness) shapes preferences, enabling precise targeting. By offering a scalable, interpretable framework, NN+S empowers firms to optimize strategies in personalized marketing, product design, and policy analysis. The remainder of this paper is organized as follows: Section 2 reviews related literature; Section 3 details the NN+S methodology; Section 4 discusses fine-tuning approaches; Section 5 presents Monte Carlo simulations; Section 6 describes the empirical application; Section 7 discusses implications; and Section 8 concludes.

2 Literature review

The broad usage of large-scale consumer data has led to the application of machine learning (ML) techniques in marketing, offering tools to model complex relationships between observable variables, such as estimating conditional expectations $F(X) = E(Y|X)$ (Paliwal and Kumar, 2009; Vellido et al., 1999). ML has been widely adopted for predictive tasks, including churn prediction (Tsai and Lu, 2009; Chen et al., 2012), sales forecasting (Kuo, 2001; Agrawal and Schorling, 1996), credit scoring (Lee and Chen, 2005; Tsai and Wu, 2008), and product return prediction (Dzyabura et al., 2023), as well as for extracting features from text (Timoshenko and Hauser, 2019).

However, these models often prioritize predictive accuracy over interpretability, limiting their utility for causal inference or structural analysis, which are critical for marketing applications like pricing or product design. Similar to this paper, [Gabel and Timoshenko \(2022\)](#) employed dimensionality reduction for product relationships. The key difference is that they employed a non-structural model, which precludes counterfactual analysis such as new product introductions or attribute changes.

In contrast, structural demand models, rooted in economic theory, provide interpretable parameters that capture consumer preferences and behavior ([Rossi et al., 1996, 2012](#)). These models typically employ a hierarchical structure where individuals choose products that maximize utility, with taste parameters varying based on demographic characteristics and purchase history ([Guadagni and Little, 1983](#)). Yet, their reliance on parametric assumptions restricts their ability to capture non-linear dependencies, prompting interest in hybrid approaches that integrate the flexibility of ML with the interpretability of structural models.

Recent studies have explored the combination of machine learning (ML) techniques with structural models for demand estimation. [Shiller et al. \(2013\)](#) studies first-degree price discrimination when firms observe rich web-browsing and related individual-level behavior alongside standard demographics. The demand system is structural yet close in form to an ordered probit, which the paper uses to quantify large simulated profit gains from behavior-based tailored pricing relative to demographic targeting alone in the empirical application. [Farrell et al. \(2020\)](#) introduced a deep learning framework for modeling observed heterogeneity in structural models, providing a theoretical foundation but offering limited practical guidance for addressing unobserved heterogeneity or relaxing the independence of irrelevant alternatives (IIA) assumption.

This paper contributes to this growing literature by introducing NN+S, a semi-parametric framework that integrates neural networks with structural models to estimate heterogeneous taste parameters. Unlike prior work, NN+S explicitly models both observed and unobserved heterogeneity, as well as intra-individual random taste shocks, using a neural network ar-

chitecture that reflects key structural components. Specifically, NN+S captures observed heterogeneity through linear and non-linear demographic effects, unobserved heterogeneity via individual uncorrelated variation and correlated variation through segment weights, and intra-individual taste shocks to account for non-IIA behavior.

This design allows NN+S to overcome the limitations of traditional structural models while retaining interpretability for causal inference. Moreover, NN+S provides practical implementation details, leveraging standard neural network tools for scalability and flexibility. By addressing the gaps in prior hybrid approaches, such as the lack of guidance for modeling unobserved heterogeneity and non-IIA individual demand, NN+S offers a more comprehensive framework for preference estimation. This is particularly valuable for marketing applications, where firms require both accurate predictions and interpretable parameters to inform strategies like personalized pricing, product recommendations, and market segmentation.

3 NN+S model

In this section, we outline the step-by-step construction of the NN+S model, incrementally incorporating its distinct neural network (NN) parts to illustrate the integration of neural networks with structural modeling principles. We draw analogies to Hierarchical Bayesian Models (HBM) to highlight the similarities and differences between the two approaches. We begin by specifying the demand model, as defined in Equation 1, before detailing the neural network components.

$$U_{ijt} = \underbrace{X_{jt} \cdot \beta_i}_{V_{ijt}} + \varepsilon_{ijt} \quad (1)$$

$$V_{ijt} = \underbrace{X_{jt}}_{1 \times K} \cdot \underbrace{\beta_i}_{K \times 1}$$

The utility of individual i purchasing alternative j at time t is modeled as the scalar product of a vector of individual-specific preferences, β_i , and a vector of product attributes,

X_{jt} , augmented by a random shock, ε_{ijt} , which follows a Type-I Extreme Value (EVT1) distribution (Gumbel distribution)¹. This specification is adopted for two reasons. First, it accommodates a range of structural interventions, including the introduction of new products, modifications to product characteristics, and shifts in individual preferences, among others. Second, the EVT1 error term yields a straightforward expression for the probability of selecting alternative j , given by the ratio of exponents in Equation 2.

$$P_{it}(j') = \frac{\exp(V_{ij't})}{\sum_j \exp(V_{ijt})} \quad (2)$$

This form aligns seamlessly with the Softmax² transformation commonly implemented in neural network frameworks, offering computational efficiency over alternative approaches while remaining a standard in structural modeling.

Given the choice probabilities for individual purchases, as defined in Equation 2, we can formulate the log-likelihood of the observed data, expressed in Equation 3. This log-likelihood is maximized with respect to the full set of neural network parameters, collectively denoted as Ω . Here, I represents the total number of individuals and T_i denotes the number of purchases made by the individual i .

$$L(\Omega) = \sum_{i=1}^I \sum_{t=1}^{T_i} \log(P_{it}(j')) \rightarrow \max_{\Omega} \quad (3)$$

Notice that in hierarchical Bayesian models (HBMs), researchers frequently use the same assumptions to obtain Equation 2³. The key difference is that NN+S provides a point estimate of β_i while HBM provides a posterior distribution of β_i . Having established the demand model, we now introduce a neural network part by specifying how individual preferences, β_i , depend on demographic characteristics, z_i .

¹See https://en.wikipedia.org/wiki/Gumbel_distribution for details.

²See https://en.wikipedia.org/wiki/Softmax_function for details on the Softmax function

³In the case of HBM, $P_{it}(j')$ is a conditional probability

$$\beta_i = \underset{K \times Z}{\Delta} \underset{Z \times 1}{z_i} + \underset{K \times 1}{G(z_i)} \quad (4)$$

As shown in Equation 4, β_i is decomposed into two distinct terms: a linear term, Δz_i , which captures the linear influence of demographic factors, and a nonlinear term, $G(z_i)$, which accounts for more complex and flexible relationships. Both parts are parameterized through a subset of trainable parameters Ω . Although we designate Δz_i as the linear part, it is worth noting that $G(z_i)$ may also exhibit a degree of linear behavior. The relative contribution of each term is governed by the strength of regularization applied to it. Specifically, a greater regularization on one term reduces its contribution to the overall model, shifting the balance between Δz_i and $G(z_i)$. The optimal regularization parameters are determined through a cross-validation procedure that scores each candidate on held-out validation folds within the training partition, as detailed in Section 4. Figure 1 illustrates the construction of a non-linear part, $G(z_i)$, within the NN+S framework. The neural network employs demographic characteristics, denoted as z_i with dimensions $Z \times 1$, as its input layer. This input is processed through a sequence of dense layers (three layers in this example) to produce a vector of preferences, $G(z_i)$, with dimensions $K \times 1$. The optimal dimensionality of the internal layers is determined by a fine-tuning procedure, which is detailed in Section 4. Although the linear part, Δz_i , might appear straightforward to implement, practical challenges arise because neural network frameworks rarely offer tools to create trainable layers that are independent of the input layers. To address this, we propose a practical workaround, as illustrated in Figure 2. This approach employs a trainable layer that technically depends on an input layer, but the input is deliberately trivial: a scalar constant, denoted as 1 with dimensions 1×1 . This input feeds into a dense layer with dimensions $K \times D$, paired with a trivial activation function⁴. This design ensures compatibility with standard neural network architectures while efficiently capturing the linear contribution to β_i . Alternatively, depending on their relative computational costs, a researcher might opt for a two-dimensional embedding

⁴ideally the simplest trainable activation, such as linear, to minimize computational cost

layer in place of the dense layer, offering a potentially more efficient solution tailored to the specific framework.

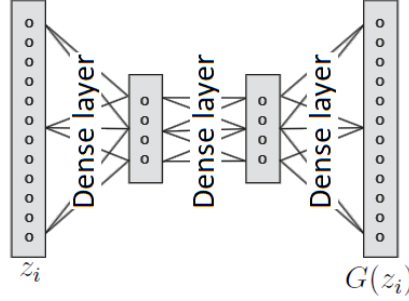


Figure 1: Definition of $G(z_i)$ through an input layer z_i and three dense layers.

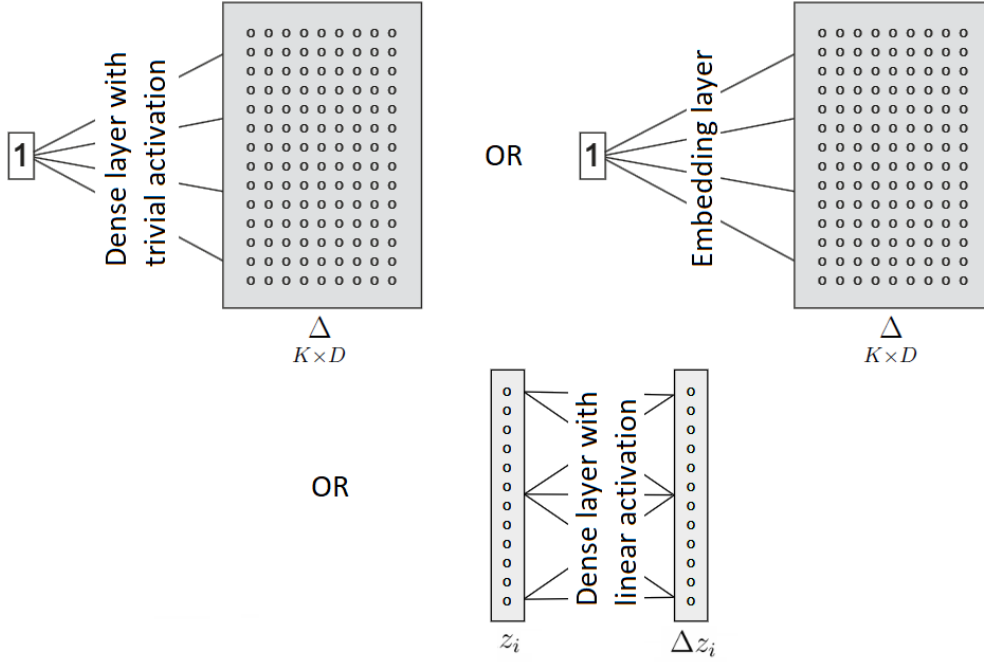


Figure 2: Definition of Δ through an input layer with constant input 1 and a two-dimensional dense layer (upper left) or an embedding layer (upper right). Alternatively, one may define Δ by transforming z_i through a dense layer with a linear activation function without the bias parameters (bottom).

Equation 4 specifies how individual preferences, β_i , depend on demographic characteristics, z_i . By observing the alternatives an individual selects, we can infer their preferences. The variation in β_i between individuals, as influenced by z_i , reflects the population-level trends captured by the combined contributions of $G(z_i)$ and Δz_i . Of course, z_i may not

fully account for all variation in β_i . Consequently, $G(z_i)$ and Δz_i are often referred to as representing *observed heterogeneity*, as this part of the variation of consumer preference can be attributed to observable individual characteristics z_i . In terms of HBM, Equation 4 looks quite the same, with two big caveats. First, there is no randomness in β_i in the NN+S model. The second caveat lies in the fact that it is difficult to estimate $G(z_i)$ using Bayesian methods because neural networks (NNs) are over-identified. Anyway, if you were able to estimate $G(z_i)$ with HBM (for example, by simplifying the functional form), but with zero variance assumption, we may say that $\beta_i \sim N(\underbrace{\Delta z_i}_{K \times Z Z \times 1} + \underbrace{G(z_i)}_{K \times 1}, 0)$. In that case, similar to the NN+S approach, there is no randomness in the individual tastes in this HBM. This slight abuse of notation will help us gain an intuition behind the other components of the NN+S model. In Equation 5, we introduce an unobservable uncorrelated part of individual heterogeneity, denoted u_i . This term, u_i , accounts for the residual variation in β_i not explained by observable factors and is commonly referred to as *unobserved heterogeneity*. It is immediately apparent that, without additional assumptions, distinguishing which portion of β_i arises from observed versus unobserved heterogeneity poses an identification challenge. To address this, we impose regularization on u_i . Consequently, the mathematical identification of the model is based on the regularization of its various parts of the NN, while the structural identification is elaborated in Section 4.

$$\beta_i = \Delta z_i + G(z_i) + u_i \quad (5)$$

To construct a term u_i using standard neural network (NN) tools, we suggest using an embedding layer with a dimensionality of K . The embedded layer is a layer that maps discrete input data, such as words or categories, into vectors of real numbers. In our setting, it is the right tool for u_i , which maps discrete input data i to a vector of real numbers $\underbrace{u_i}_{K \times 1}$. Figure 3 illustrates the construction of u_i with NN.

Equation 5 can be compared to HBM with $\beta_i \sim N(\underbrace{\Delta z_i}_{K \times Z Z \times 1} + \underbrace{G(z_i)}_{K \times 1}, \frac{1}{\alpha_u} K \times K)$, where α_u is a scalar and I is a diagonal matrix. One of the key differences between the NN +

S and HBM approaches is that α_u is estimated in HBM. α_u is responsible for the size of the unobserved heterogeneity. In other words, the functional form of the distribution β_i and the observed data provide the identification for α_u and, as a result, for the size of the unobserved heterogeneity. In the NN+S approach, the size of unobserved heterogeneity is defined by the regularization, which is identified by a cross-validation procedure. In other words, the functional form of regularization and the observed data provide the identification for u_i regularization through a cross-validation procedure and, as a result, for the size of the unobserved heterogeneity. We claim that the fact that NN+S relies on the cross-validation procedure to identify the size of the unobserved heterogeneity may provide superior predictive performance by separating the functional form assumption from the size of the unobserved heterogeneity.

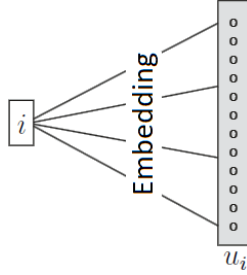


Figure 3: Definition of u_i through an input layer i and an embedding layer.

So far, the identification of β_i for individual i is as follows: the systematic variation in taste parameters β_i with respect to demographic characteristics z_i identifies the observed heterogeneity part $\Delta z_i + G(z_i)$. The residual deviations from this systematic variation identify the unobserved heterogeneity part u_i , with the relative contributions of the observed and unobserved heterogeneity determined by cross-validation. By construction, the components of the vector u_i are independent, meaning that if an individual i exhibits a preference for a particular attribute, such as "organic," that deviates from what is predicted by z_i , this provides no information about their preferences for other attributes, such as "free range" or "all natural." In scenarios with an infinitely long purchase history, the independent components of u_i would suffice to fully estimate the unobserved taste preferences. However, with finite

purchase data, the model cannot capture potential correlations between preferences for related attributes such as "organic," "free range," and "all natural." To address this limitation, we introduce a reduced dimensionality part v_i , as presented in Equation 6, which captures correlated unobserved heterogeneity by modeling shared preference structures across individuals.

$$\beta_i = \Delta z_i + G(z_i) + u_i + \underset{K \times R R \times 1}{A} v_i \quad (6)$$

v_i can be viewed as the weights that the individual i has in R groups of tastes (or market segments). A is a matrix composed of R column vectors of group tastes. For example, group 1 may represent preferences for "organic", "free range", "all natural", etc., while group 2 may represent preferences related to sugar content, fruit flavor, or any other aspect of sweetness. As you may notice that part Av_i is similar to part Δz_i , this fact highlights the fact that v_i can be interpreted as some unobserved demographic characteristics for researchers, such as individual health, sugar addiction, or any other individual characteristics that can influence the heterogeneity of unobserved preferences. The dimensionality reduction part v_i can be constructed similarly to u_i , but with an extra dense layer representing A , which is shown in Figure 4. In this case, v_i components should be constrained to make sure that they are always nonnegative and add up to 1, as in Equation 16 (left plot in Figure 4). Alternatively, one may consider introducing a Softmax layer after the embedding layer (right plot). In terms of the HBMs, the correlated unobservable heterogeneity can be defined through $\beta_i \sim N(\underset{K \times Z Z \times 1}{\Delta} z_i + \underset{K \times 1}{G}(z_i), \Omega_u)$, where Ω_u is any proper variance-covariance matrix. To make a comparable analogy, our dimensionality reduction is similar to imposing a dimensionality reduction for the matrix $\Omega_u = A \cdot A'$.

We have now arrived at the final part of the NN+S model, which requires us to make a slight change in notation. So far, we have been building an individual demand function with

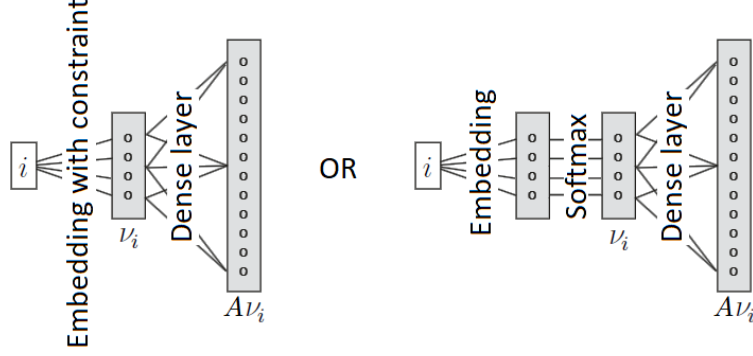


Figure 4: Definition of v_i through an input layer i , an embedding layer, and a dense layer.

I.I.A.⁵ property, which is widely considered unrealistic⁶. There are several ways to relax the I.I.A. assumption; however, in this paper, we propose a specific solution that can be easily implemented using NN tools. We introduce intra-individual correlated random shocks in taste parameters. Each individual i with the probability $\frac{1}{M}$ receives a shock μ^m to their taste vector β_i . $m \in [1, \dots, M]$ is randomly drawn each time an individual makes a purchase. Naturally, the value of μ is identified from all deviations from the I.I.A. demand.

$$\beta_i^m = \beta_i + \mu^m \quad (7)$$

$$\beta_i^m = \Delta z_i + G(z_i) + u_i + A v_i + \mu^m \quad (8)$$

$$U_{ijt}^m = \underbrace{X_{jt} \cdot \beta_i^m}_{V_{ijt}^m} + \varepsilon_{ijt} \quad (9)$$

$$V_{ijt}^m = \underbrace{X_{jt}}_{1 \times K} \cdot \underbrace{\beta_i^m}_{K \times 1} \quad (10)$$

$$P_{it}^m(j') = \frac{\exp(V_{ij't}^m)}{\sum_j \exp(V_{ijt}^m)} \quad (11)$$

⁵Independence of irrelevant alternatives

⁶Note that still, our individual I.I.A. demands produce non-I.I.A. demand on the aggregate level.

$$P_{it}(j') = \sum_{m=1}^M \frac{P_{it}^m(j')}{M} \quad (12)$$

Let's compare our approach to HBM. In some situations, researchers assume intra-individual heterogeneity in the form of a random vector with normal distribution $\beta_{it} \sim N(\bar{\beta}_i, \Omega_\mu)$, where $\bar{\beta}_i \sim N(\Delta z_i + G(z_i), \Omega_u)$ (Krueger et al. (2021), Allenby and Rossi (1998)). Potentially, we could have imposed a normal distribution assumption as well and attempted to calculate the probabilities by approximating it with discrete points, but this would have been computationally burdensome. Our approach requires fewer calculations of the probabilities in Equation 11, but potentially allows far more degrees of freedom.

$$\beta_{it} = \begin{cases} \Delta z_i + G(z_i) + u_i + Av_i + \mu^1 \text{ with probability } \frac{1}{M} \\ \Delta z_i + G(z_i) + u_i + Av_i + \mu^2 \text{ with probability } \frac{1}{M} \\ \dots \\ \Delta z_i + G(z_i) + u_i + Av_i + \mu^M \text{ with probability } \frac{1}{M} \end{cases} \quad (13)$$

As you may see, this approach allows a non-I.I.A. demand (Equation 12) on the level of making individual decisions, while keeping I.I.A. on the level of each separate m (Equation 11), which allows us to efficiently calculate $P_{it}^m(j')$ using Softmax and calculate $P_{it}(j')$ through a collapse of the corresponding dimension by averaging, which is also called in the NN context a "reduction operation" or "average pooling".

u_i and v_i are identified only for existing customers, so it is important to discuss how to make predictions for new customers. We want to highlight the three most prominent solutions. 1) Estimate the distribution of u_i and v_i for existing customers and assume that a new customer is drawn from the same distribution. 2) Use the specification β_i^m without u_i and v_i (see Equation 14) for both estimation and prediction. 3) Assume that u_i and v_i are zero for new customers. The major drawback of solutions 1 and 3 is that the estimated model differs from the one used for the prediction, or it requires multiple computations of

the final probability to calculate the expectation. Solution 2 utilizes the same model and calculation procedure for both training and prediction, which is why this paper will focus on that solution.

$$\beta_i^m = \Delta z_i + G(z_i) + \mu^m \quad (14)$$

Notice that the interpretation and identification of μ^m term in Equation 14 is different from Equation 8. Because Equation 14 does not include unobserved individual heterogeneity, μ^m takes the role of aggregating unobserved individual heterogeneity. In other words, in this specification, μ^m accounts for both the distribution of unobserved individual heterogeneity and non-I.I.A. behavior through intra-individual random variation. Implementing μ is similar to implementing Δ because it does not depend on the input data. Figure 5 illustrates a fragment of the NN used to create μ .

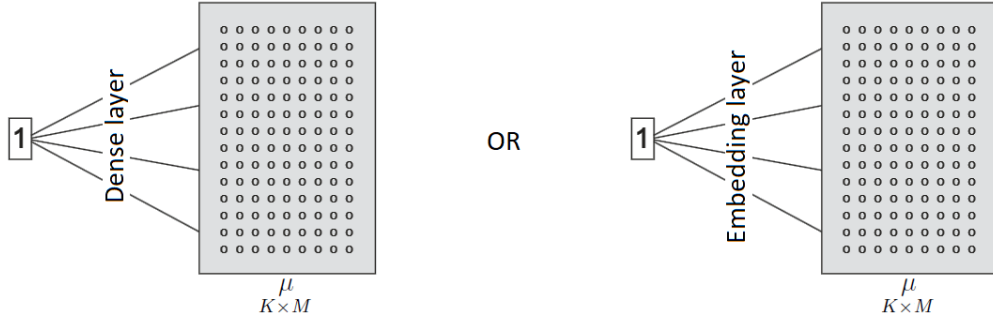


Figure 5: Definition of μ through a dense layer or an embedding layer and a constant input.

Notice that after introducing all these structural NN parts, we have to rewrite the target function to account for all regularizations.

$$L(\Omega) = \sum_{i=1}^I \sum_{t=1}^{T_i} \log(P_{it}(j')) - \lambda_{\Delta} \sum_{k=1}^K \sum_{d=1}^D (\Delta_{kd})^2 - \lambda_G \sum_{w=1} (W_{G,w})^2 - \lambda_u \sum_{i=1}^I \sum_{k=1}^K (u_{ik})^2 - \lambda_A \sum_{k=1}^K \sum_{r=1}^R (A_{kr})^2 - \lambda_{\mu} \sum_{k=1}^K \sum_{m=1}^M (\mu_k^m)^2 \rightarrow \max_{\Omega} \quad (15)$$

$$\sum_r v_{ir} = 1, \text{ for each } i, v_{ir} > 0 \text{ for each } i \text{ and each } r \quad (16)$$

Equation 16 records the simplex restrictions on v_i from Figure 4 in algebraic form. Here, λ_Δ , λ_G , λ_u , λ_A , and λ_μ are nonnegative penalty weights for the squared L_2 penalties on Δ , on the trainable weights defining $G(\cdot)$, on the embeddings $\{u_i\}$, on A , and on the mass points $\{\mu^m\}$, as in Equation 15. They determine how aggressively each block is shrunk toward zero, reallocating fit across the structured parts of the model. We select these weights jointly with the other architecture choices using the cross-validation procedure in Section 4.

4 Two approaches to the fine-tuning (cross-validation)

In this section, we discuss best in-sample (BIS) and best out-of-sample (BOS) approaches to cross-validation (i.e. fine-tuning) of the parameters of the NN+S model. We begin with an introduction to the general problem, followed by a detailed description of the two approaches. In our opinion, researchers often pay insufficient attention to the question of fitting noise versus fitting the trend in structural models. When a researcher assumes certain functional forms, she imposes constraints that may prevent the model from accurately identifying the true ratio between the noise and the trend. For example, in a hierarchical individual demand model, assuming a specific functional form for individual preferences based on demographic characteristics (observed heterogeneity) and a normal distribution for unobserved heterogeneity may lead to underfitting or overfitting of the noise (unobserved heterogeneity in this case). The neural network (NN) framework suggests choosing hyperparameters (including regularization strengths) by k -fold cross-validation. In standard practice, each candidate specification is scored on held-out validation folds constructed inside the training partition, so that folding mimics predictive accuracy while keeping a separate test partition available for final evaluation after all tuning choices are fixed. However, in our case, individual purchase occasions are not fully independent due to individual unobserved heterogeneity.

Cross-validation remains applicable provided each fold respects that dependence structure and the data-generating process (DGP) is comparable across the training and validation portions of the folds. We propose two distinct cross-validation designs depending on whether the DGP supplies new purchases for the same individuals (in-sample) or new individuals (out-of-sample).

The best in-sample (BIS) approach is used when the goal is predictions for the same individuals as in the estimation window. In BIS, the training and validation portions of each fold contain the same consumers (typically on disjoint purchase occasions), so the procedure selects hyperparameters that score well on future purchases by those individuals. In contrast, the best out-of-sample (BOS) approach targets predictions for new individuals. In BOS, the training and validation portions of each fold contain disjoint sets of consumers, so the procedure selects hyperparameters that score well when past purchases are unavailable at prediction time. To illustrate the intuition behind BIS and BOS, consider how each approach determines the balance between observed and unobserved heterogeneity. BIS addresses the question: Given the previous purchase history of these individuals, how much can demographic characteristics further explain their preferences, thereby optimizing predictions of future purchases? BOS, on the other hand, addresses the following questions: given only demographic characteristics, how much can these characteristics explain preferences for new individuals? Consequently, BOS relies more heavily on demographic characteristics than BIS. The choice between BIS and BOS for a particular structural estimate depends on the researcher’s assumptions. If the researcher assumes a representative sample of the population, BIS may be appropriate, with the understanding that the estimated observed heterogeneity provides additional precision based on previous observations. If the researcher aims to predict preferences for different individuals, BOS should be used.

Table 1: Table of model parameters

Part	Cross-validated	Estimated
Δz_i	Δ regularization coefficient λ_Δ	Matrix elements Δ_{kd}
$G(z_i)$	$G(.)$ regularization coefficient λ_G	Layers' weights $W_{G,w}$
	$G(.)$ number of layers	
	$G(.)$ activation function	
u_i	u_i regularization coefficient λ_u	Vectors u_i $K \times 1$
Av_i	A regularization coefficient λ_A	Matrix elements A_{kr}
		Vectors v_i $R \times 1$
μ^m	μ^m regularization coefficient λ_μ	Vectors μ^m $K \times 1$

5 Monte Carlo Simulations

This section evaluates the NN+S approach's ability to recover the true taste parameters and offers insights into the various parts of the model.

5.1 Recovery of Observed Heterogeneity

Suppose a researcher aims to recover the functional form of how taste parameters vary across the population based on demographic characteristics z_i . A naive two-step approach might involve estimating individual taste parameters $\hat{\beta}_i$ first and then regressing these estimates on z_i . The NN+S approach offers a superior one-step estimation method that leverages variation in demographic characteristics for identification. To illustrate this, we present an example where each individual has only one observation, making the two-step approach infeasible, while NN+S successfully estimates the parameters. This exercise demonstrates that NN+S utilizes cross-sectional variation in purchasing patterns based on demographic characteristics. The simulated data consist of $I = 10000$ individuals and $J = 5$ products, each with two product characteristics. Figure 6 displays the distribution of the simulated demographic characteristics. Each individual i has two demographic characteristics z_{1i} and

z_{2i} , drawn from a uniform distribution within a circular domain.

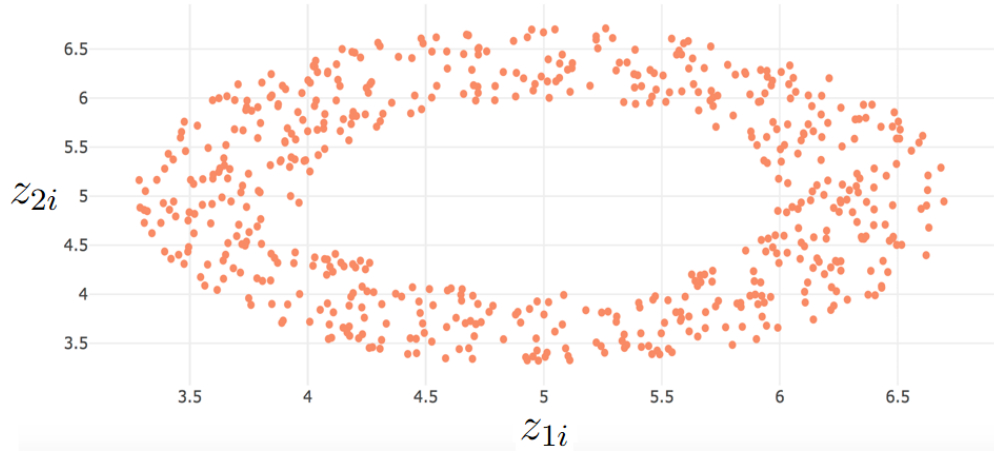


Figure 6: Distribution of MC simulated individual demographic characteristics z_i : z_{1i} and z_{2i}

True individual taste parameters are generated from demographic characteristics according to the following equations and scaled to a range of 1 to 4 (as shown in the upper two graphs of Figure 7):

$$\begin{aligned}\beta_{1i} &= (5 - z_{1i})^2 + (5 - z_{2i})^2 \\ \beta_{2i} &= (10 - z_{1i})^2 + z_{2i}^2\end{aligned}\tag{17}$$

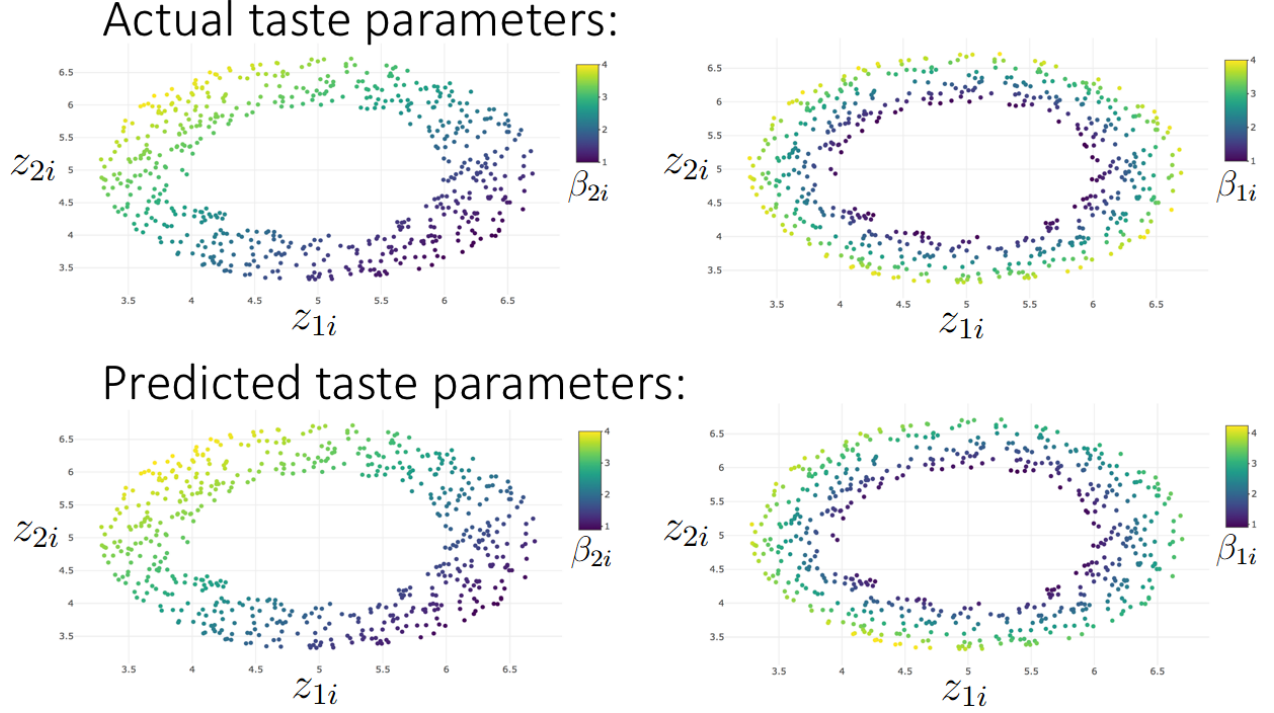
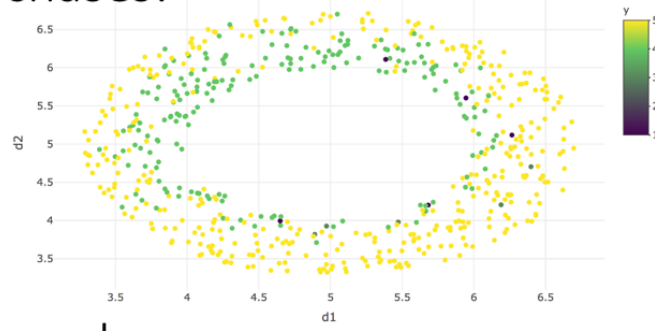


Figure 7: Taste parameters: true and recovered by NN+S

Figure 7 illustrates how the NN+S approach estimates the relationship between taste parameters β_i and demographic characteristics z_i . The top two graphs display the ground-truth taste parameters, while the bottom two graphs show the taste parameters estimated by the NN+S model. Each point represents an individual with specific demographic characteristics z_{1i} and z_{2i} , with the color indicating the value of their taste parameters β_{1i} and β_{2i} . Despite the neural network being misspecified⁷, the model correctly estimates the dependency between demographic characteristics and taste parameters.

⁷The neural network is misspecified because the true taste parameters are quadratic functions of the demographic characteristics.

Actual purchases:



Predicted purchases:

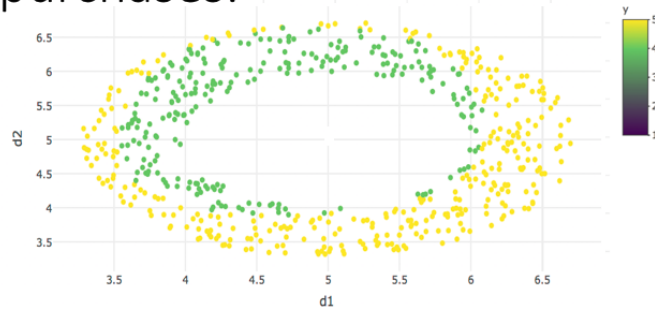


Figure 8: Purchases: true and recovered by NN+S

Figure 8 compares actual and predicted purchases, visually representing the alternative chosen (indicated by color) for each individual. As the figure demonstrates, the NN+S approach accurately predicts the buying behavior of most consumers.

5.2 Recovery of Unobserved Heterogeneity

To illustrate the properties of the parts of the model responsible for the unobserved heterogeneity, we conducted a series of Monte Carlo simulations. For all simulations in this Section, we assume $I = 300$ individuals, where each individual has $T_i \in [1, 100]$ purchase occasions. During each purchase occasion $t = 1..T_i$, the individual i can choose one of the 8 alternative products. Each product has 10 attributes, whose characteristics are drawn from a standard distribution $X_{ijt} \sim N(0, 1)$ ⁸. The distribution of T_i is skewed toward lower numbers to ensure a sufficient number of individuals with poorly defined preferences. Figure 9

⁸This assumption of random draws for each purchase occasion is unrealistic but allows for fewer observations. The results are generalizable to more realistic distributions of X_{ijt} .

shows the distribution of the number of individual purchases.

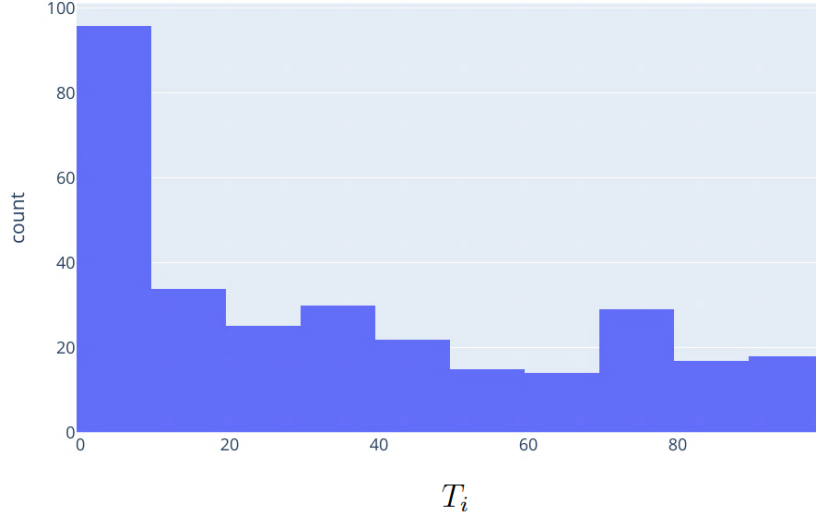


Figure 9: Distribution of the number of individual purchases.

Table 2 presents the results of the estimation of three different data-generating processes (DGP) and three different model specifications. The root mean squared error (RMSE)⁹ is reported for two groups of customers: those with $T_i \in [1, 10)$ and those with $T_i \in [50, 100]$. Equations 18, 19, and 20 define the three DGPs: individual variation only, group variation only, and both group and individual variations. Equations 21, 22, and 23 define the three model specifications used to evaluate the behavior of terms u_i and Av_i .

$$\beta_i = b_0 + u_i \quad (18)$$

$$\beta_i = \begin{cases} b_0 + b_1, & \text{with probability 0.5} \\ b_0 + b_2, & \text{with probability 0.5} \end{cases}, \text{ where } \begin{aligned} b_0 &= (1, -2, 0, -1, -2, 0, -1, 0, 0, 0)' \\ b_1 &= (0, 0, 0, 0, 0, 1, -1, 0, 1, -1)' \\ b_2 &= (0, 0, 0, 1, -1, 0, 0, 1, -1, 0)' \end{aligned} \quad (19)$$

⁹If β_i^* is the vector of true preferences and $\hat{\beta}_i$ is the vector of estimated preferences, then $\text{RMSE} = \sqrt{\sum_i \frac{1}{I \cdot K} (\beta_{ik}^* - \hat{\beta}_{ik})^2}$.

$$\beta_i = \begin{cases} b_0 + b_1 + u_i, & \text{with probability 0.5} \\ b_0 + b_2 + u_i, & \text{with probability 0.5} \end{cases} \quad (20)$$

, where u_i and $u_{ik} \sim N(0, 1)$, $b_0 = (1, -2, 0, -1, -2, 0, -1, 0, 0, 0)'$, $b_1 = (0, 0, 0, 0, 0, 1, -1, 0, 1, -1)'$, $b_2 = (0, 0, 0, 1, -1, 0, 0, 1, -1, 0)'$

$$\beta_i = \beta_0 + u_i \quad (21)$$

$$\beta_i = \beta_0 + Av_i \quad (22)$$

$$\beta_i = \beta_0 + u_i + Av_i \quad (23)$$

Table 2: Monte Carlo simulations results for unobservable heterogeneity part

Data Generating Process	Specification	RMSE of taste preferences	
		$T_i \in [1, 10)$	$T_i \in [50, 100]$
Only group variation	$\beta_i = \beta_0 + u_i$	0.457	0.284
	$\beta_i = \beta_0 + Av_i$	0.352	0.098
	$\beta_i = \beta_0 + u_i + Av_i$	0.349	0.098
Only individual variation	$\beta_i = \beta_0 + u_i$	0.550	0.286
	$\beta_i = \beta_0 + Av_i$	0.604	0.538
	$\beta_i = \beta_0 + u_i + Av_i$	0.550	0.286
Group and individual variations	$\beta_i = \beta_0 + u_i$	0.372	0.265
	$\beta_i = \beta_0 + Av_i$	0.370	0.290
	$\beta_i = \beta_0 + u_i + Av_i$	0.367	0.227

Table 2 reports the results of the Monte Carlo simulations. For each data-generating process (DGP) and each model specification, we selected the regularization penalty by k -fold cross-validation using held-out validation folds within the simulated estimation sample. We refitted each specification with the selected penalty and calculated the root mean squared

error (RMSE) of the taste parameters.

First, consider the DGP with only group variation. In theory, using only u_i should suffice to estimate unobserved heterogeneity with an infinite sample of purchases. However, because we have a finite sample and u_i is high-dimensional, it struggles to estimate taste parameters as precisely as the group unobserved heterogeneity part Av_i . Second, for the DGP with only individual variation, Av_i is too low-dimensional to account for the true individual unobserved heterogeneity. As a result, it struggles to provide precise estimates, even as the number of purchase occasions increases. Third, for the DGP with both group and individual variations, having only u_i should theoretically be sufficient to estimate unobserved heterogeneity with an infinite sample. However, in practice, with a finite sample, the u_i part alone provides low performance. Similarly, Av_i cannot account for all the variation. Naturally, the combination $u_i + Av_i$ provides the best estimates by leveraging both individual and group-level variations.

6 Empirical application

In this section, we apply the NN+S approach to a real-world dataset of individual purchases to assess the model’s performance and demonstrate its practical utility. We compare the NN+S model to benchmark models, including a linear hierarchical Bayesian model (LHBM) and the model proposed by [Farrell et al. \(2020\)](#) (FLM), using both in-sample and out-of-sample prediction metrics. The empirical analysis highlights the contributions of different parts of the NN+S model and provides insights into its strengths and limitations in applied settings. Since u_i and v_i , which represent individual-specific effects and segment weights, respectively, cannot be identified for individuals without prior purchase observations, we employ different model specifications for in-sample and out-of-sample datasets, as shown in Equations [24](#) and [25](#).

$$\beta_i^m = \Delta z_i + G(z_i) + u_i + Av_i + \mu^m \quad (24)$$

$$\beta_i^m = \Delta z_i + G(z_i) + \mu^m \quad (25)$$

Equation 25 excludes u_i and v_i , relying solely on demographic effects and intra-individual taste shocks for out-of-sample predictions. This ensures robust estimation for new consumers, where unobserved heterogeneity cannot be inferred from purchase histories.

6.1 Data

We use a dataset of margarine purchases from a scanner panel [Allenby and Rossi \(1991\)](#), a well-known dataset in the marketing field. This dataset includes 4,470 purchases made by 516 consumers across 10 different brands. It also contains demographic information for each consumer (e.g., income, family size, education) and the price of each brand during each shopping trip. The dataset’s combination of purchase histories, demographic characteristics, and product attributes makes it ideal for testing the NN+S model’s ability to capture both observed and unobserved heterogeneity. The descriptive statistics of the original dataset are presented in Tables 3 and 4. To evaluate the model’s performance for both existing and new customers, we divide the data into a training partition and a test partition using two approaches:

- **In-sample split:** For each consumer, purchase occasions are randomly split between the training partition and the test partition, so every consumer appears in the test partition with at least some purchase occasions retained in the training partition (Figure 10).
- **Out-of-sample split:** Consumers are randomly assigned to either the training partition or the test partition, with no consumer appearing in both (Figure 11).

These splits assess predictive accuracy for the same individuals (in-sample) and for new individuals (out-of-sample). Hyperparameters (including regularization strengths) are selected

by k -fold cross-validation on the training partition only: each held-out fold within that partition serves as a validation fold for scoring candidate specifications. The test partition does not enter any cross-validation fold used for selection. After choosing hyperparameters, we refit each model on the full training partition and report hit rates on the test partition. This yields four labeled subsamples in total. Two under the in-sample construction, purchase sets in the training and test partitions. Two under the out-of-sample construction, consumer groups in the training and test partitions.

Table 3: Descriptive statistics of 10 margarine brands.

Brand	1	2	3	4	5	6	7	8	9	10
Market share	39.5%	15.6%	5.4%	13.2%	7.0%	1.6%	7.1%	4.5%	5.0%	0.7%
Mean price	0.518	0.543	1.015	0.437	0.345	0.781	0.825	1.077	1.189	0.569
S.d. price	0.151	0.120	0.043	0.119	0.035	0.115	0.061	0.030	0.014	0.072

Table 4: Descriptive statistics of 516 consumers.

Statistic	Mean	St. Dev.	Min	25%	75%	Max
Number of trips	8.663	6.624	1	4	11	43
Income	27.558	17.052	2.5	17.5	32.5	130
Family Size	2.922	1.275	1	2	4	8
College Education	0.300	0.459	0	0	1	1
White Collar	0.581	0.494	0	0	1	1
Retired	0.231	0.422	0	0	0	1

Based on 516 consumers

For each consumer, their individual purchase history was randomly split between the training partition and the test partition, ensuring that any consumer represented in the test partition has at least some purchase occasions in the training partition. The resulting subsamples are referred to as the in-sample construction.

Each consumer is randomly assigned to either the training partition or the test partition, ensuring that there are no consumers shared between the two partitions. The resulting subsamples are referred to as the out-of-sample construction.

Customer ID	Demographics Zi	Product characteristics X	Purchases Y											
			t=1	t=2	t=3	t=4	t=5	t=6	t=7	t=8	t=9	t=10	t=11	t=12
2100016	...	...	1	1	1	1	1	4	1					
2100024	...	...	1	4	1	1	4	8	4	3	1	1	1	
2100495	...	...	1	1	1	1	1	1	1	1	1	1	1	
2100560	...	...	2	4										
2100610	...	...	7	4										
2100628	...	...	1											
2100685	...	...	1	2	3	4	3	3						
2100693	...	...	9	9	9	4	9	9	4	9	3	3	9	3

Customer ID	Demographics Zi	Product characteristics X	Purchases Y					
			t=1	t=2	t=3	t=4	t=5	t=6
2100016	...	...	1	1	4	1		
2100024	...	...	4	1	3	1	1	
2100495	...	...	1	1	1	1	1	
2100560	...	...	4					
2100610	...	...	7					
2100685	...	...	2	3				
2100693	...	...	9	9	4	4	9	9

Customer ID	Demographics Zi	Product characteristics X	Purchases Y					
			t=1	t=2	t=3	t=4	t=5	t=6
2100016	...	...	1	1	1			
2100024	...	...	1	1	4	8	4	1
2100495	...	...	1	1	1	1	1	1
2100560	...	...	2					
2100610	...	...	4					
2100685	...	...	1	4	3	3		
2100693	...	...	9	9	9	3	3	3

Training partition

Test partition

Figure 10: In-sample construction (BIS): purchase occasions split across the training and test partitions. Hyperparameters are tuned using validation folds within the training partition only following similar split procedure within the training partition.

Customer ID	Demographics Zi	Product characteristics X	Purchases Y											
			t=1	t=2	t=3	t=4	t=5	t=6	t=7	t=8	t=9	t=10	t=11	t=12
2100016	...	...	1	1	1	1	1	4	1					
2100024	...	...	1	4	1	1	4	8	4	3	1	1	1	
2100495	...	...	1	1	1	1	1	1	1	1	1	1	1	
2100560	...	...	2	4										
2100610	...	...	7	4										
2100628	...	...	1											
2100685	...	...	1	2	3	4	3	3						
2100693	...	...	9	9	9	4	9	9	4	9	3	3	9	3

Customer ID	Demographics Zi	Product characteristics X	Purchases Y											
			t=1	t=2	t=3	t=4	t=5	t=6	t=7	t=8	t=9	t=10	t=11	t=12
2100016	...	...	1	1	1	1	1	4	1					
2100495	...	...	1	1	1	1	1	1	1	1	1	1	1	
2100628	...	...	1											
2100693	...	...	9	9	9	4	9	9	4	9	3	3	9	3

Customer ID	Demographics Zi	Product characteristics X	Purchases Y										
			t=1	t=2	t=3	t=4	t=5	t=6	t=7	t=8	t=9	t=10	t=11
2100024	...	...	1	4	1	1	4	8	4	3	1	1	1
2100560	...	...	2	4									
2100610	...	...	7	4									
2100685	...	...	1	2	3	4	3	3					

Training consumers

Test consumers

Figure 11: Out-of-sample construction (BOS): disjoint sets of consumers in the training and test partitions. Hyperparameters are tuned using validation folds within the training partition only following similar split procedure within the training partition.

6.2 Results

We evaluated the performance of the NN+S model using hit rates on the test partitions defined above. Table 5 presents the hit rates for different specifications under the out-of-sample construction (new customers), while Table 6 reports the hit rates under the in-sample construction. The results show that the full NN+S model, incorporating demographic effects (Δz_i and $G(z_i)$), unobserved heterogeneity (u_i and Av_i), and intra-individual taste shocks

(μ^m), achieves the highest hit rate for in-sample predictions (1354 out of 2149 observations). This outperforms reduced specifications, such as those omitting unobserved heterogeneity or non-linear demographic effects. For out-of-sample predictions, the full model achieves a hit rate of 1,016 out of 2,254 observations, slightly surpassing simpler specifications but falling short of some benchmarks, such as the linear hierarchical Bayesian model (LHBM), as noted elsewhere in the study. These findings underscore the value of capturing unobserved heterogeneity and non-IIA (Independence of Irrelevant Alternatives) behavior, especially when purchase histories are available. For marketing applications, this capability enables personalized strategies, such as tailored promotions or product recommendations, by leveraging individual-specific and segment-level insights. However, for new customers with limited data, simpler models may perform adequately if demographic effects are primarily linear.

Table 5: The hit rate results for different specifications of the model for out-of-sample datasets (new customers).

Specification	Comments	# Hits	Hit rate
$\beta_i^m = \Delta z_i + G(z_i) + \mu^m$	Full model for BOS	1016	45.1%
$\beta_i^m = \Delta z_i + G(z_i)$	No μ^m (FLM)	998	44.3%
$\beta_i^m = \Delta z_i + \mu^m$	No $G(z_i)$	1009	44.8%
$\beta_i^m = G(z_i) + \mu^m$	No Δz_i	1013	44.9%
$\beta_i^m = \beta_0 + \mu^m$	No demographics	1005	44.6%
$\beta_i^m \sim \text{Mixture of Normals}$	(LHBM)	1001	44.4%

* Out of 2254 observations

7 Discussion

The empirical application of the NN+S model to the margarine dataset serves as an illustrative example of the model’s capabilities, importance of BIS vs BOS fine-tuning, and as an illustration of the roles of the different parts of the model. This dataset was selected for its prominence in the marketing literature, serving as a well-established benchmark for comparing demand estimation models. Despite the simplicity of the demographic variables

Table 6: The hit rate results for different specifications of the model for in-sample datasets (same customers, different trips).

Specification	Comments	# Hits	Hit rate
$\beta_i^m = \Delta z_i + G(z_i) + u_i + Av_i + \mu^m$	Full model for BIS	1354	63.0%
$\beta_i^m = \Delta z_i + G(z_i) + u_i + Av_i$	No μ^m	1342	62.4%
$\beta_i^m = \Delta z_i + u_i + Av_i + \mu^m$	No $G(z_i)$	1343	62.5%
$\beta_i^m = G(z_i) + u_i + Av_i + \mu^m$	No Δz_i	1341	62.4%
$\beta_i^m = \Delta z_i + G(z_i) + Av_i + \mu^m$	No u_i	1237	57.6%
$\beta_i^m = \Delta z_i + G(z_i) + u_i + \mu^m$	No Av_i	1325	61.7%
$\beta_i^m = \beta_0 + u_i + Av_i + \mu^m$	No demographics	1330	61.9%
$\beta_i^m = \Delta z_i + G(z_i) + \mu^m$	No unobserved heterogeneity	997	46.4%
$\beta_i^m = \Delta z_i + G(z_i)$	Only $G(z_i) + \Delta z_i$ (FLM)	993	46.2%
$\beta_i^m \sim \text{Mixture of Normals}$	(LHBM)	1227	57.1%

* Out of 2149 observations

available (e.g., income, family size), the NN+S model demonstrates its ability to extract additional predictive power, achieving higher in-sample hit rates compared to alternative approaches like LHBM and FLM. This suggests that even in datasets with basic demographic information, NN+S can leverage unobserved heterogeneity and purchase histories to improve predictions. However, the true potential of NN+S is likely to be realized in more complex settings with richer individual-level data, such as detailed consumer demographics or behavioral traits. In such environments, the model’s ability to capture non-linear relationships and correlated unobserved heterogeneity through segment weights would be even more pronounced. Additionally, while traditional LHBM tools, such as those implemented in the `bayesm` package, may struggle with the computational demands of larger datasets or more complex models, the NN+S framework benefits from the rapid advancements in neural network infrastructure (`TensorFlow`, `PyTorch`, `Keras`), which continues to improve in terms of speed and scalability.

The margarine dataset also highlights the critical role of fine-tuning through cross-validation in optimizing the NN+S model’s performance. While cross-validation is essential for balancing the model’s flexibility and avoiding overfitting, it introduces complexities in interpreting the structural parameters. For instance, the choice between best in-sample (BIS)

and best out-of-sample (BOS) fine-tuning procedures affects the balance between fitting observed and unobserved heterogeneity, complicating the structural interpretation of the results. This trade-off is not unique to NN+S; it is a broader challenge in structural modeling, where functional form assumptions implicitly resolve the balance between fitting the trend and the noise. In the case of LHBM, cross-validation using the BIS approach suggested an optimal number of components as five, while the BOS approach indicated only one component, underscoring the hidden trade-off in traditional models as well. Further research is needed to better understand how fine-tuning impacts the interpretability of structural parameters in hybrid models like NN+S.

Looking ahead, the NN+S framework opens up exciting possibilities for integrating new types of data into structural models, leveraging the flexibility of neural networks to handle diverse data formats. For example, product images, audio recordings (e.g., voice reviews), and text data (e.g., product reviews or social media posts) could be incorporated as inputs to the neural network, providing richer information about consumer preferences and product attributes. Such data are increasingly available through e-commerce platforms, social networks, and other digital channels, offering a wealth of unstructured information that traditional structural models struggle to accommodate. By including these data types into the NN+S architecture, researchers and practitioners could gain deeper insights into consumer behavior, enabling more accurate demand estimation and more effective marketing strategies. Future research should explore these extensions, investigating how multimodal data can be seamlessly integrated into structural frameworks to enhance both predictive power and interpretability.

8 Conclusion

This paper introduces NN+S, a novel semi-parametric framework that integrates neural networks with structural models to estimate heterogeneous taste parameters with enhanced

accuracy and interpretability. By embedding a neural network architecture that explicitly models observed heterogeneity, unobserved heterogeneity, and intra-individual random taste shocks, NN+S captures complex preference structures while retaining the structural interpretability necessary for causal inference. This approach bridges a critical gap in the literature, providing practical guidelines for implementing neural networks in structural modeling, particularly for addressing unobserved heterogeneity and non-IIA behavior, areas where prior work offered limited guidance.

Monte Carlo simulations demonstrate that NN+S effectively estimates the taste parameters. The empirical application to a margarine purchase dataset shows that NN+S outperforms benchmark models, such as linear hierarchical Bayesian models (LHBM) and the model by [Farrell et al. \(2020\)](#) (FLM) in in-sample predictions, achieving a hit rate of 63% compared to 57% and 46%, respectively. This improvement is driven by NN+S’s ability to leverage purchase histories through individual-specific effects and segment weights, offering firms a powerful tool for personalized marketing strategies, such as targeted pricing and product recommendations.

For out-of-sample predictions, NN+S showed a marginal improvement over the benchmarks, with hit rates of 45.1%, 44.4%, and 44.3%, respectively.

The NN+S model offers several practical advantages for marketers and policymakers. Its flexibility in modeling non-linear relationships and unobserved heterogeneity enables more accurate demand estimation, particularly in data-rich environments. The model’s scalability, facilitated by standard neural network tools, makes it suitable for large-scale applications, while its interpretability ensures that insights can be translated into actionable strategies, such as optimizing pricing or product design.

Future research could extend NN+S by incorporating multimodal data (e.g., product images, reviews) or exploring context-specific cross-validation to further enhance its predictive power and applicability to diverse marketing contexts.

In summary, NN+S offers a flexible and interpretable framework that advances the inte-

gration of machine learning and structural modeling, providing a valuable tool for preference estimation and decision-making in marketing and beyond.

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